OpenClaw AI Weather Station Tutorial

> This article assumes that the Jetson Nano Docker environment, OpenClaw expansion board, DHT, OLED, RGB, and three dialogue methods are already set up. This article only covers the AI Weather Station application.
>
> Features in this tutorial may require training. If command control fails, you can first describe the operation process in detail, then refine the dialogue.

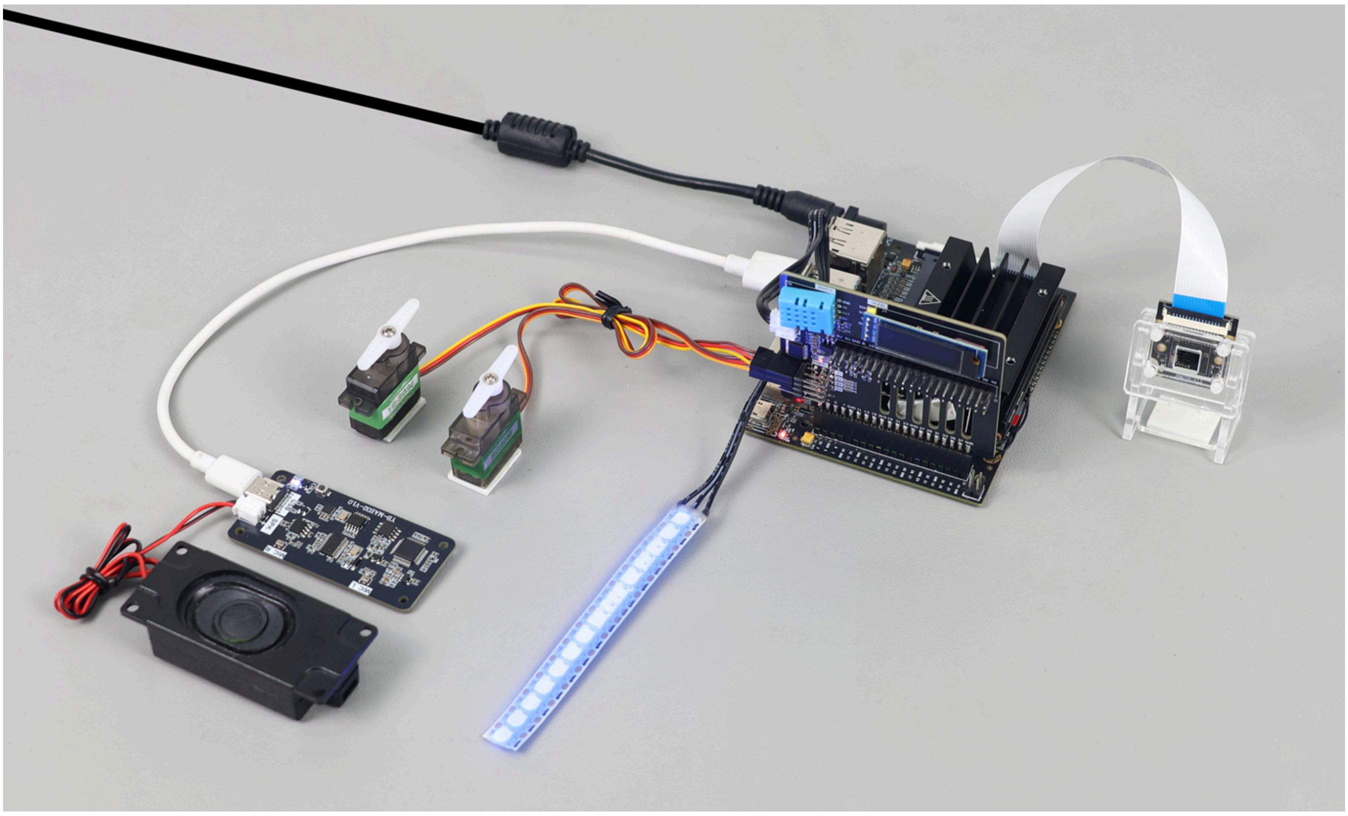
1. Tutorial Objectives

The AI Weather Station combines "temperature and humidity reading + OLED display + RGB status light + natural language broadcast" to implement a desktop environment status assistant. After completion, you can:

- Get current temperature and humidity.
- Display temperature and humidity or weather card on OLED.
- RGB shows different colors based on temperature status.
- Provide a brief evaluation of environmental comfort.
- Support TUI, WhatsApp, and voice triggers.

2. Hardware Preparation

- Jetson Nano B01
- OpenClaw expansion board
- RGB light strip
- Optional: AI voice interaction module + speaker
- Wiring diagram



3. Software Entry

Enter the Jetson Docker container:

```
~/run_usb_docker.sh
```

Common commands:

```
/usr/local/bin/dht --format json
/usr/local/bin/dht_oled --format json
/usr/local/bin/rgb --format json color --color blue
/usr/local/bin/oled text --text "weather"
```

Recommended temperature status rules:

- Temperature low: RGB lights blue.
- Temperature comfortable: RGB lights yellow.
- Temperature high: RGB lights red.

4. Basic Capability Testing

Enter TUI:

```
openclaw tui
```

```
root@jetson-yahboom:~# openclaw tui

 OpenClaw 2026.3.8 (3caab92) — I'll do the boring stuff while you dramatically stare at the logs like it's cinema.

 openclaw tui - ws://127.0.0.1:18789 - agent main - session main

 session agent:main:main
 gateway connected | idle
 agent main | session main (openclaw-tui) | bailian/qwen-plus-2025-09-11 | tokens 0/200k (0%)
```

You can test first:

- Read current temperature and humidity and display on OLED
- Change RGB light to green
- Control both servos to rotate to 180 degrees

5. Three Dialogue Methods

Method	Suitable Scenario	Recommended Use
TUI	Local debugging of DHT and OLED	Prioritize verifying sensor chain
WhatsApp	Remote query of indoor environment	Suitable for viewing results on phone
Voice	Desktop broadcast	Suitable for one-sentence trigger of weather station

5.1 WhatsApp Method

You can send:

- What is the current indoor temperature and humidity
- AI Weather Station
- How is the indoor environment now? If it's too hot, remind me

20:34

+86 153 3885 7526 (你)
给自己发消息

20:33

How is the indoor environment now? If it is too hot, remind me. 20:34

Current indoor environment readings:

🌡️ **Temperature:** 29.7°C (85.5°F)
💧 **Humidity:** 54%

Assessment: The temperature is **just below the hot threshold** (30°C). It's warm but not excessively hot yet.

Recommendation:

- The room is comfortably warm, but if the temperature rises above 30°C, I'll remind you with a red light warning
- Current humidity at 54% is in a comfortable range (40-60% is ideal)
- Consider ventilation if you feel warm or if temperature continues to rise

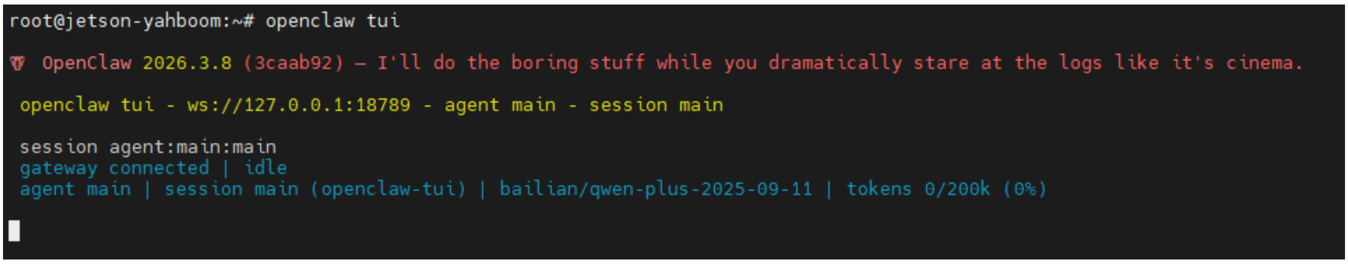
Would you like me to set up a reminder if the temperature exceeds 30°C?

Note: I did not schedule a reminder in this turn, so this will not trigger automatically. 20:34

5.2 TUI Method

Run:

```
openclaw tui
```



Recommended inputs:

- AI Weather Station
- Read temperature and humidity, light red if temperature is high, light yellow if temperature is comfortable
- Display temperature and humidity on OLED, and indicate temperature status with RGB

5.3 Voice Method

Execute the following command while openclaw gateway is running:

```
cd /root/yahboom_ws/src/largemodel
python3 main.py
```

You can say:

- AI Weather Station
- How is the current indoor environment
- Tell me if it's too hot and stuffy

```
root@jetson-yahboom:~/yahboom_ws/src/largemodel# python3 main.py
[main] Jetson OpenClaw voice launcher
[main] bridge ready
[main] tts ready
[main] asr ready
[main] Voice interaction is running. Press Ctrl+C to stop.

[asr] I'm here

[user] AI气象站。
[assistant] 🌦AI 气象站已更新
            📊 当前环境：温度 28.9 度，湿度 58%
            📟OLED：已显示温湿度
            🌈 RGB：已切换为黄色呼吸灯，用来提示当前温度状态
```

6. Comprehensive Cases

6.1 Environmental Sampling

Experiment objective: Read current indoor temperature and humidity.

Example commands:

- Read current temperature and humidity
- What is the current indoor temperature
- Help me check the current humidity

6.2 Comfort Judgment

Experiment objective: Make simple judgments based on temperature and humidity.

Example commands:

- Help me judge if the current environment is comfortable
- Is the current temperature and humidity suitable for prolonged office work
- Give me a brief evaluation based on temperature and humidity

6.3 Stuffy Reminder

Experiment objective: Read temperature and humidity, then judge if it's stuffy.

Example commands:

- Tell me if it's too hot and stuffy
- Is it a bit stuffy now
- Based on temperature and humidity, judge if ventilation is needed now

6.4 OLED and RGB Linkage

Experiment objective: Display environment status on OLED, and indicate temperature status with RGB.

Example commands:

- AI Weather Station
- Display temperature and humidity on OLED, and indicate temperature status with RGB
- Low temperature lights blue, comfortable temperature lights yellow, high temperature lights red

7. Common Problems

- Temperature and humidity reading failed: First execute `dht` to verify.
- OLED not displaying: First execute `dht_oled` for individual testing.
- Only replying with numbers without evaluation: Need to confirm whether comfort and stuffy judgment logic is added in local route or bridge.
- RGB not changing: Check if the reply generated an RGB action.